MANGO: An Autodiff Neutrino Oscillation Engine

for End-to-End Differentiable Analysis Pipelines

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MANGO: A Neutrino Gradient Oscillator

1. The Derivative Gap & JAX-Native Engine

Native JAX Transformation Graph:

- `jax.jit`: Trace compilation for zero overhead on GPU & CPU.
- `jax.vmap`: Vectorized batching across $(E, \cos\theta_z)$ grids.
- `jax.grad / vjp`: Exact reverse AD with $O(1)$ parameter scaling.
- `jax.tree_util`: PyTrees native to `NumPyro` & `BlackJAX`.

- **The Derivative Gap**: Legacy engines (`OscProb`, `Prob3++`, `NuFast`) evaluate $P_{\alpha\beta}$ in forward mode only. Computing sensitivities requires $2N_{\text{params}}$ **finite difference passes** with severe step-size tuning instabilities across rapid oscillation phases.
- **The MANGO Paradigm**: Written from scratch in pure JAX, evaluating **exact machine-precision sensitivities** w.r.t **all** Hamiltonian, trajectory, and detector parameters in a **single reverse pass**.
- **Differentiable Spherical Chord Metric**: Trajectory through **43 PREM shells** parameterized in closed form:

$$s(r) = \sqrt{r^2 - r_{\min}^2}, \quad r_{\min} = (R_\oplus - d_{\det})\sqrt{1-\cos^2\theta_z}$$

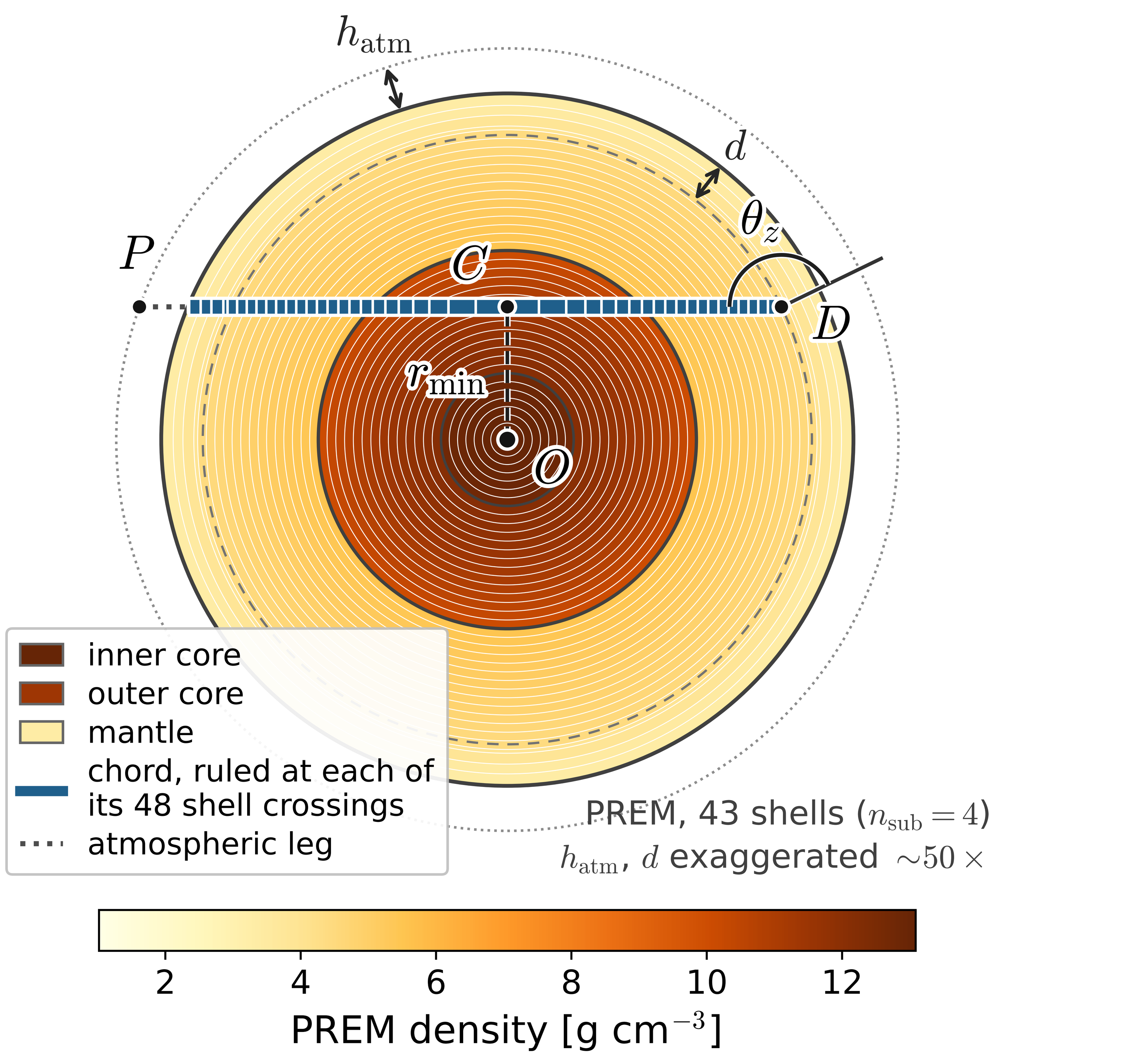

Figure 1: Chord geometry through 43 PREM shells ($n_{\text{sub}} = 4$, 48 segments). Path lengths $\{L_k\}$ are **smooth, closed-form differentiable functions**.

- **Singularity-Free Divided Differences**: Evaluates $\exp(-iHL)$ without explicit diagonalization $\longrightarrow$ **eliminates** $\frac{1}{\lambda_i - \lambda_j}$ **singularities at MSW degeneracies.**
- **Sub-linear Reverse-Mode Scaling**: Output-dimension scaling ($O(1)$ in input count):

Derivative Target	Parameters	Overhead vs. Forward
Standard Oscillation Parameters	6	2.5×
Spherical Trajectory Geometry	42	2.9×
Layered PREM Earth (Densities, Y_e, Radii)	369	**3.0×** (**250× faster than FD**)

Multi-Tier Numerical Validation Benchmarks:

- **Forward Accuracy**: $< 10^{-8}$ vs `OscProb/NuFast` (6×10^{-15} vs numerical MSW)
- **AD Gradient Precision**: $< 10^{-9}$ relative error vs centered finite differences
- **Physical Limits Recovery**: Bit-for-bit Standard Model recovery (10^{-15})

2. Atmospheric PREM Derivative Landscapes

Unified Multi-Channel Jacobian in a Single `jacfwd` Pass:

$$J(E, \cos\theta_z) = \nabla_{\{\theta_{\text{osc}}, \boldsymbol{\rho}_{\text{Earth}}, \boldsymbol{g}_{\text{geom}}, \boldsymbol{\varepsilon}_{\text{BSM}}\}} P_{\mu e}(E, \cos\theta_z)$$

Earth Density Tomography:

$\frac{\partial P}{\partial \ln \rho_{\text{core}}}$ & $\frac{\partial P}{\partial \ln \rho_{\text{mantle}}}$ provide exact Jacobians for Earth interior inversion, with an abrupt core shadow cutoff at $\cos\theta_z \leq -0.838$.

Non-Analytic Geometry AD:

$\frac{\partial P}{\partial \cos\theta_z}$ & $\frac{\partial P}{\partial h_{\text{atm}}}$ differentiate spherical chord lengths $\{L_k\}$ and production height, reaching steep extrema of ± 40.

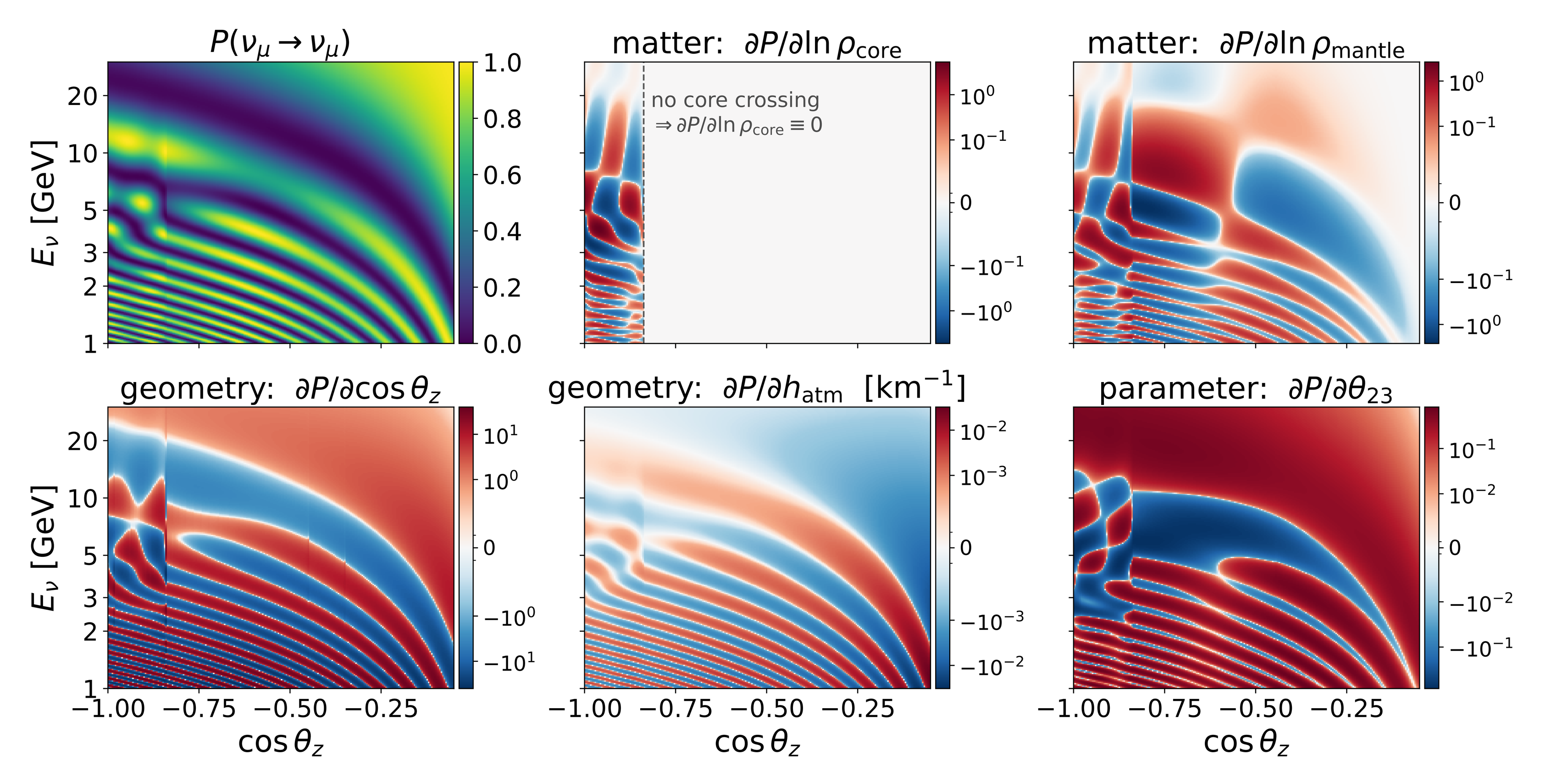

Figure 2: Complete 5-channel sensitivity landscape across PREM Earth on a 220×220 grid ($E_\nu \in [1, 30]$ GeV). Top: forward $P(\nu_\mu \to \nu_e)$ and standard θ_{23} sensitivity. Bottom: logarithmic core & mantle density gradients alongside non-analytic zenith $\cos\theta_z$ and production height h_{atm} derivatives.

3. Solar MSW & Non-Analytic BSM Derivatives

Continuous Solar Matter & MSW Resonance Gradients:

Exact functional sensitivity kernels $|\frac{\delta P_{ee}}{\delta \ln N_{e(r)}}|$ across 80 SSM shells, tracking the sharp MSW peak at $E = 4.41$ MeV without numerical derivative singularities.

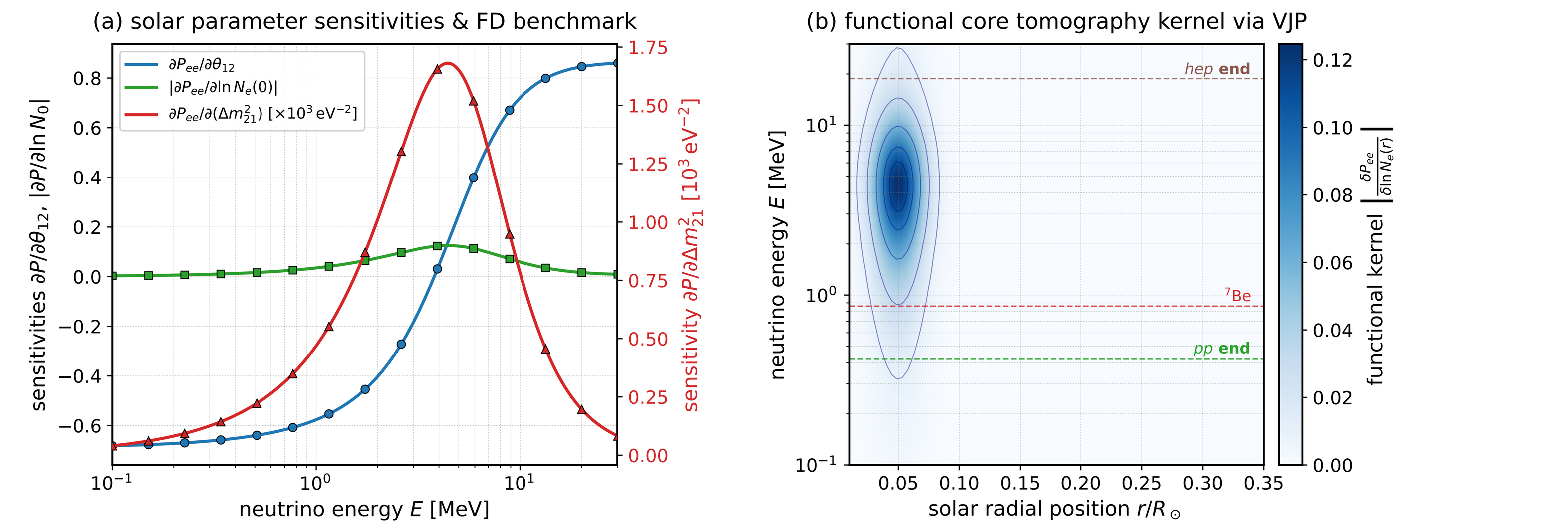

Figure 3: Solar neutrino AD. **(a)** Sharp MSW resonance peak in $\frac{\partial P_{ee}}{\partial(\Delta m^2_{21})}$ (1.68×10^3 eV^{-2}). **(b)** 2D continuous solar core tomography functional sensitivity kernel across 80 shells (1.6 s).

Beyond-the-Standard-Model Sensitivities at the SM Limit:

Instant machine-precision Jacobians for Lindblad environmental decoherence (γ_{ij}) and non-unitary mixing (α_{ij}) directly at the SM point ($\gamma = 0, \alpha = 0$) for zero-derivation profile likelihoods.

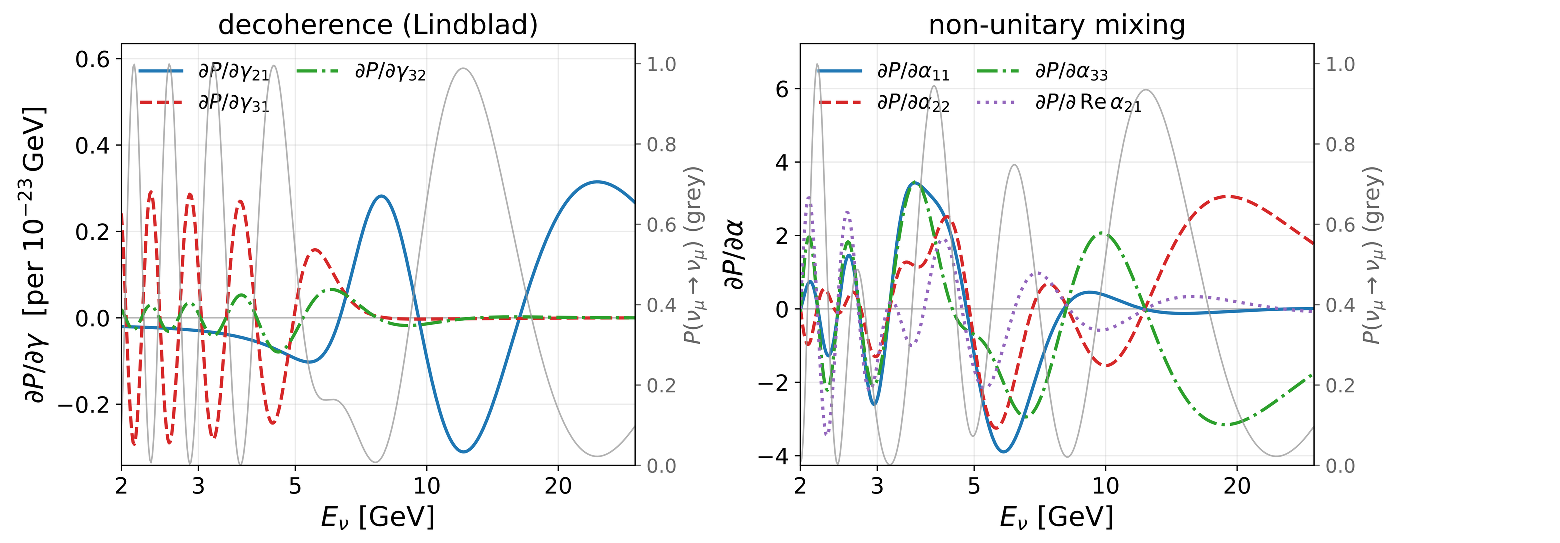

Figure 4: Sensitivities at SM point ($\gamma = 0, \alpha = 0$): **(a)** Lindblad decoherence $\frac{\partial P}{\partial \gamma_{ij}}$. **(b)** Non-unitary mixing $\frac{\partial P}{\partial \alpha_{ij}}$ through PREM Earth at $\cos\theta_z = -1$.

4. Using the Derivatives: Earth Interior Inversion

$$\underbrace{H(\boldsymbol{\vartheta}) \longrightarrow P_{\alpha\beta}}_{\text{1. Oscillation Physics}} \longrightarrow \underbrace{\mu_b \longrightarrow F_{ij} \longrightarrow F^{-1}}_{\text{2. Fisher Inversion}} \longrightarrow \underbrace{\sigma_i = \sqrt{(F^{-1})_{ii}} \longrightarrow \frac{\partial \sigma_i}{\partial \theta_{\text{sys}}}}_{\text{3. Sensitivity Reach}}$$

Engine Inputs:

- $\boldsymbol{\vartheta}$: Oscillation & PREM density
- $P_{\alpha\beta}$: Transition probability

Event Statistics:

- μ_b: Binned counts $(E_b, \cos\theta_z)$
- F_{ij}: Fisher information matrix

Inference & Reach:

- F^{-1}: Covariance matrix
- σ_i: Error & $\frac{\partial \sigma_i}{\partial \theta_{\text{sys}}}$ reach

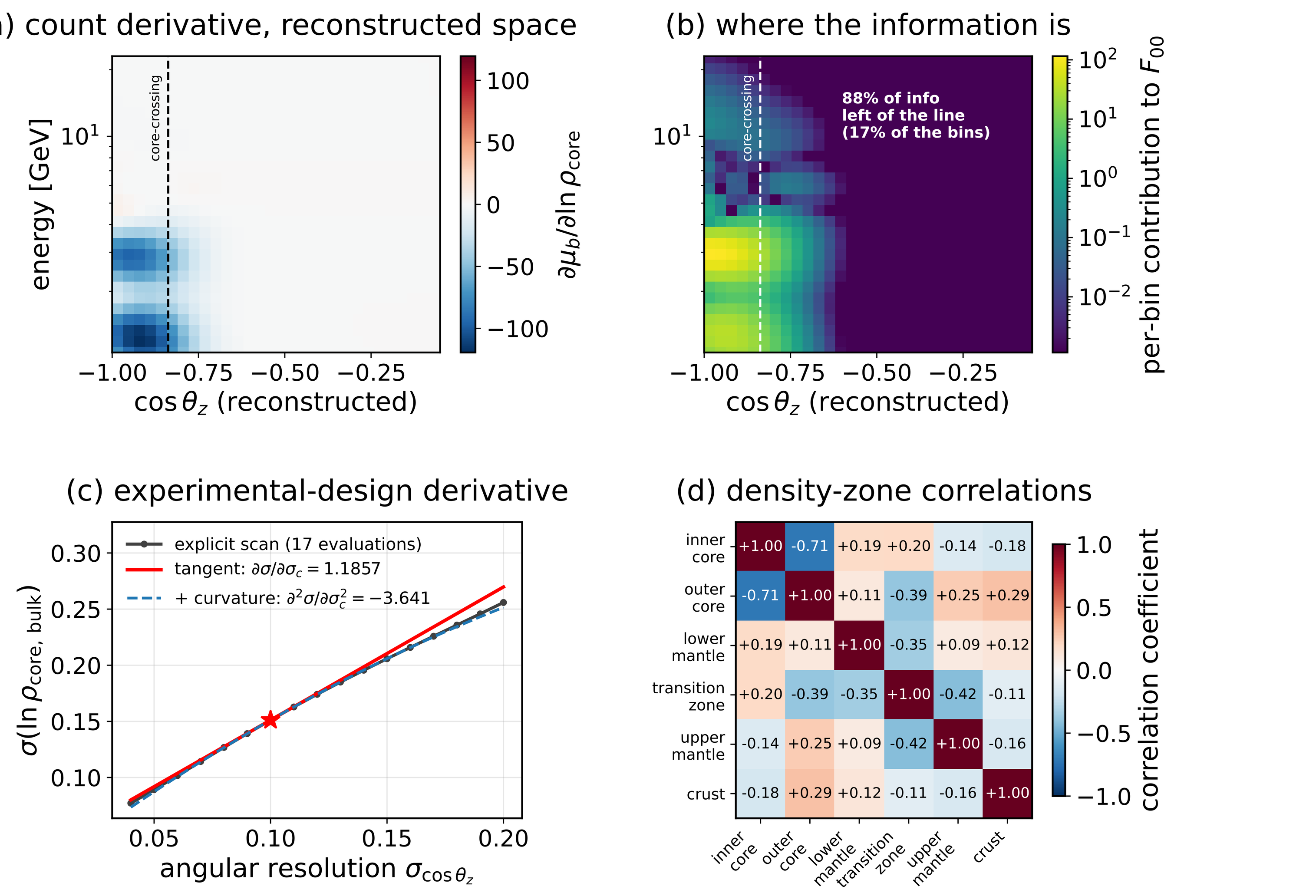

Figure 5: 11-parameter Earth interior fit (1.5×10^5 events). **(a)** Reconstructed count derivatives. **(b)** Fisher information localized below core threshold. **(c)** Instant sensitivity gradients $\frac{\partial \sigma}{\partial \sigma_{\text{det}}}$ via nested AD vs. 17 forward grid scans. **(d)** Full 11-param covariance matrix.

Principle: Hardware Co-Design via Nested AD `grad(jacfwd)` through F^{-1}:

- **Differentiating Through Matrix Inversion**: Propagates exact sensitivities through the Fisher matrix and its inverse $(F^{-1})_{ii}$, bypassing brute-force re-simulations.
- **Instant Hardware Gradients**: Evaluates analytical sensitivity response $\frac{\partial \sigma}{\partial \theta_{\text{detector}}}$ and local curvature $\frac{\partial^2 \sigma}{\partial \theta^2_{\text{detector}}}$ in a single backward pass.
- **End-to-End Experiment Optimization**: Enables gradient-based co-optimization of detector resolution, binning schemes, and baselines directly against physics discovery reach.

5. Outlooks & Differentiable Ecosystem

3D Earth Tomography & Geophysics

Functional sensitivity kernels $\frac{\delta P}{\delta \rho(r)}$ enable gradient-based 3D inversion of lateral mantle heterogeneities and Large Low-Shear-Velocity Provinces (LLSVPs).

Scalable Bayesian Inference (HMC/NUTS)

Native JAX PyTrees connect directly to Hamiltonian Monte Carlo for exact gradient sampling over $100+$ systematic nuisance parameters without grid scans.

Automated Optimization & ML Surrogates

Algorithmic co-optimization of binning grids, baseline distances, and detector response for maximum Fisher sensitivity, alongside fast neural surrogate training.

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Open Source Code

github.com/pgranger23/mango-osc